Nutri Scan AI: Depth-Assisted Nutritional Estimation Using RGB-D Food Images and Monocular Depth Learning

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Abstract**—**Estimating nutrition details from food images is difficult because each food item has a unique appearance and size, which is challenging to measure and largely depends on the environment or angle of acquisition. Calorie counting is traditionally done manually which is error-prone and tedious. Here, in this project we introduce Nutri Scan AI, an end-to-end depth assisted deep learning framework for automatic nutrition estimation from RGB images of foods along with additional depth information. We train a CNN-based model with RGB modality features along with encoded depth as input to predict calories and macros directly from an image of food. We use Nutrition5k RGB-D dataset which contains food images along with depth maps and nutrition labels as ground truth for supervised training. Additionally, we also show results with monocular depth estimation with MiDaS to predict depth from a single RGB image. We also show that depth prediction from just RGB images can be fused with original image modalities to aid in nutrition estimation.

Keywords—machine learning, deep learning, RGB-D learning, nutrition estimation, food image analysis, calorie prediction, multimodal learning, convolutional neural networks, Nutrition5k dataset

GitHub Link:
https://github.com/22SriVarsha/ML_Project

I. INTRODUCTION

Automatic estimation of nutrition through food imagery is receiving increasing attention in computer vision, healthcare, and smart meal monitoring applications. Traditional methods of calorie counting rely on manual food logging, nutritional databases, barcode scanning or user's estimation of serving sizes. However, these methods are tedious, often inaccurate and not standardized between users. Nutrition estimation from food images is difficult since food can vary drastically in color, texture, geometry, illumination, and portion size presented. RGB images also lack depth and volume data which poses problems for portion-size analysis. As calories are derived from the amount and volume of food consumed, making predictions with only 2D-image features can lead to significant error.

Deep learning approaches in computer vision have allowed convolutional neural networks, transfer learning, and multimodal learning to be applied towards automating food-recognition and nutrition-analysis algorithms. However, most prior works attempt to classify food or recognize ingredients while neglecting geometry such as thickness, height and volume. Depth-assisted learning has recently been explored, allowing models to gain information about food structure and portion size which cannot be easily obtained from RGB images. Most RGB-D nutrition systems require specialized depth sensors and acquisition settings hindering real world application. Depth estimation using a monocular RGB image has also been less widely studied.
Contributions of this paper are listed below:

- We introduce Nutri Scan AI, an RGB-depth multimodal learning framework for AI-based automatic nutrition estimation from food photography.
- We introduce depth assisted learning for effective portion-size comprehension and enhanced nutrition prediction accuracy.
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- We benchmark our proposed framework with Nutrition5k RGB-D dataset with deep learning based regression techniques.
- We leverage monocular depth estimation with MiDaS to provide support for RGB-only deployments without requiring any hardware depth sensor.

II. RELATED WORK

Food recognition and nutrition estimation are one of the most popular topics in computer vision, due to the need of smart dietary monitoring systems. Early studies relied on hand-engineered image features and traditional machine learning based classifiers to classify food. Since the rise of deep learning, CNN networks such as AlexNet, VGGNet, ResNet and EfficientNet achieve state-of-the-art food recognition performance by learning deep image representations from food images. Supervised deep learning approaches perform well on publicly

available food datasets for food classification tasks. Although deep learning-based models are extensively explored in food classification, only a few models estimate food nutrition (such as calories, carbs, protein and fat).

Nevertheless, food portion size and nutrition estimation from a single RGB image remains challenging. Most of the previous approaches either don't reason about geometry or require specific acquisition settings and hardware. Furthermore, there are very few that use RGB feature learning, depth learning, monocular depth estimation and nutrition regression. Unlike other methods, Nutri Scan AI system combines RGB-depth learned features with monocular depth to support better automatic nutrition estimation, and future deployment of RGB-only applications.

III. METHODOLOGY

A. Problem Formulation

We defined the nutritional estimation problem as multimodal regression problem where the target is to directly predict nutrition from food images and depth data. We selected Mean Absolute Error (MAE) as our loss function as nutrition estimation is a regression problem with continuous numeric values and the MAE loss function produces easily interpreted values of prediction error.

RGB images provide useful information about food appearance - texture, color, and overall appearance. Besides appearance, depth maps also provide geometric information about the food such as its thickness, height and size of the food portion. Because our method is based on RGB and depth information, our model extracts joint representations about the food based on appearance and geometry information which is useful in estimating food nutritional content compared to RGB images alone.

B. Model Architecture

Our Nutri Scan AI system is made up of two streams: an RGB feature extraction stream and a depth feature extraction stream. Our RGB feature extraction branch uses a pretrained convolutional neural network (CNN) ResNet18 - pretrained on the ImageNet dataset. We selected ResNet18 as our CNNs because it has been shown to learn the best features from images while being relatively cheap and easily transferable to our regression problem. We added our own fully connected regression layers to replace the final linear classification layer of ResNet18 to predict nutritional values.

We also created a network for extracting features from normalized depth maps, which can be obtained from the Nutrition5k RGB-D dataset or by estimating depth from our monocular network, which has a few convolutional layers followed by some non-linear activation and adaptive pooling layers. The depth encoder network extracts geometric information of the food such as food structure, surface texture and depth of the food in the image.

With our RGB feature vector and depth feature vector from each branch, we combine the two vectors with a fusion technique. We then feed the representation vector to fully connected regression layers to get our nutritional output. These include energy, food weight, fat, carbs, and protein. Through multimodal fusion of both RGB and depth data, we learn from both appearance and geometry features of the food.

IV. EXPERIMENTAL SETUP

A. Datasets

For this project, we used a large-scale RGB-D food dataset called Nutrition5k. This dataset provides RGB-D images for the applications of nutrition estimation and food analysis. The dataset consists of ~5k food dishes taken by Intel RealSense depth cameras, which includes aligned RGB food images, raw depth maps, and nutritional annotations. In addition to calories, each sample is annotated with information about the food's mass, carbohydrates, protein, and fat content as well as ingredient-level metadata. We chose this dataset because unlike most image food datasets, this dataset provides both visual appearance information along with depth information. The depth information provides insight about the geometry of the food which is directly related to portion size.

We utilized overhead RGB food images along with the corresponding raw depth images as inputs to our proposed framework. Nutrition5k also comes with official train/test split files which we used during experimentation. During preprocessing, we removed any dishes that were missing either a valid rgb image or depth image. Then, we resized all rgb images and depth maps to be 224×224 pixels in resolution before inputting them into our deep learning model. Since the depth maps provided by Nutrition5k are raw RealSense values, we convert them into floating-

point normalized depths using the scaling factor provided with the dataset.

Nutrition5k consists of many different types of food with variation in texture, shape, color, serving size, and arrangement. This allowed us to see how well our model performed in realistic food-analysis settings and allowed for enough variation to properly train multimodal nutritional prediction models.

TABLE I. ***Summary of Datasets***

Dataset	Train	Test	Total Samples	Input
Nutrition5k	80%	20%	5000 dishes	RGB Images, Depth Maps

B. Evaluation Metrics

To evaluate the results of the proposed Nutri Scan AI framework, we used regression metrics as nutrition prediction is a regression problem where the target variables are continuous numeric values such as calories, food mass, fat, carbohydrates and protein. Our main evaluation metric in the experiments is Mean Absolute Error (MAE) which is the average absolute difference between the predicted nutrition labels and ground-truth labels. MAE is a widely adopted regression metric that's easily interpretable as the unit is the same as the target attribute.

We conducted qualitative analysis using RGB images, depth-map visualization and 3D surface visualizations from predicted depth maps. We further studied the food geometry, thickness, and portion sizes found in the model. To understand the impact of various depth-based features such as mean depth, maximum depth, minimum depth, depth range and relative volume score, we conducted an ablation study using calorie prediction.

C. Implementation Details

We implemented the Nutri Scan AI system in Python and using various machine learning libraries such as Pytorch, Torchvision, OpenCV, NumPy, Pandas and Matplotlib. We used a pre-trained ResNet18 model with pre-trained ImageNet weights for our RGB feature extraction to perform transfer learning and speed up the training process. We replaced the final classification layer with fully connected regression layers to predict the nutrition values.

Our depth encoder consists of multiple convolutional layers with nonlinear activations and adaptive pooling. We combined the features extracted from these with RGB and depth data and then used regression layers to predict calories and macronutrients. We predict nutritional values {calories, food mass, fat, carbohydrates, protein}.

In addition to Nutrition5k depth maps, we also predicted monocular depth maps using MiDaS pretrained model from Torch Hub. MiDaS enables us to estimate depth maps from RGB food images automatically without using a depth sensor. We also visualize depth maps as grayscale depth maps and as 3D surface plots to show the shape and size of food.

V. RESULTS

Data Distribution and Statistical Analysis

Before we train our network on Nutrition5k, we took a look at the distributions of nutritional information available in the dataset to understand what sort of data the network will learn from. Fig. 1 shows histogram of the ground truth calorie counts across Nutrition5k. We can see from the figure that while most dishes have low-to-moderate calorie counts, some dishes have extremely high calorie counts as well. The dataset is therefore skewed to higher calorie counts with some extreme outliers. Datasets with nutritional information from real world diets often suffer from this problem because there exist certain combinations of foods with very large servings sizes which yield unnaturally high nutrition values.

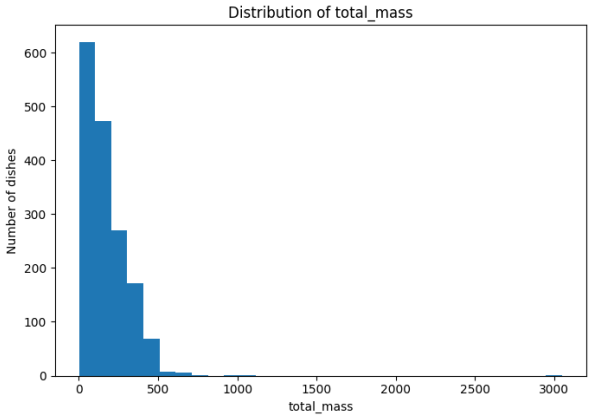


Fig 1: Distribution of total mass

The correlations between each pair of variables were also analysed. As you can see from the correlation matrix above, calories seem to be closely tied to many other nutritional attributes. Total calories are strongly correlated with both total fat (correlation of 0.82) and total protein (correlation of 0.81). This makes sense as fat and protein content would be a good predictor of how calorific a meal/snack is. It also makes sense that total calories are strongly correlated with total mass (correlation of 0.77) as larger portions will contain more nutritional value.

However, you can see that total carbohydrates have a weaker correlation to total fat (correlation of 0.19). This isn't surprising as foods high in carbs do not necessarily contain high fat.

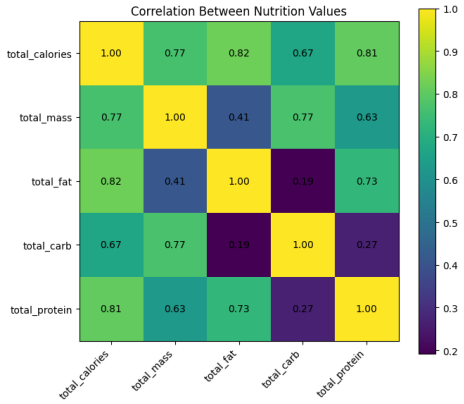
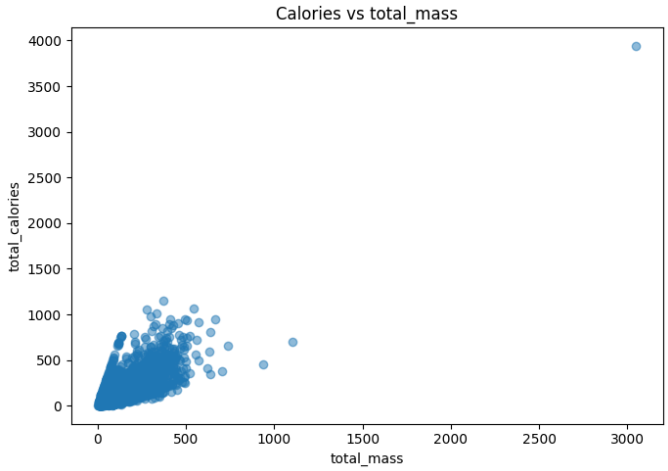
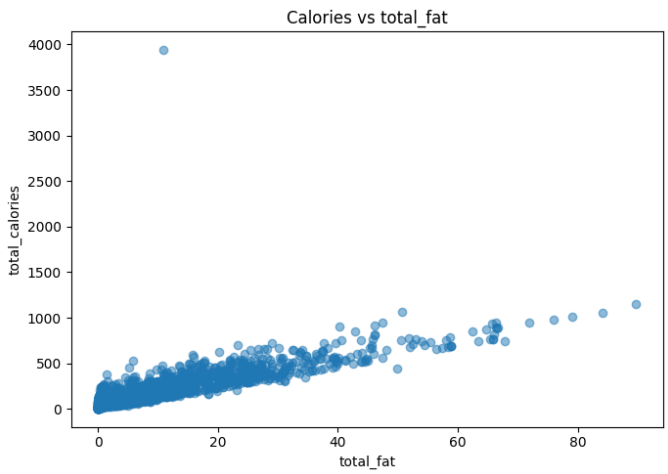


Fig 2: Correlation between Nutrition values






```
Depth Feature Data:
          dish_id  mean_depth  min_depth  max_depth  depth_range  \
0  dish_1561662216    0.370307     0.2796     6.5535       6.2739
1  dish_1561662054    0.365620     0.3022     1.1530       0.8508
2  dish_1562008979    0.363888     0.2633     5.0114       4.7481
3  dish_1560455030    0.436043     0.2974     1.9239       1.6265
4  dish_1558372433    0.394667     0.2993     0.4546       0.1553

   relative_volume_score  total_calories  total_mass  total_fat  total_carb  \
0           90483.523438      300.794281       193.0  12.387489   28.218290
1           93617.742188      419.438782       292.0  23.838249   26.351543
2           91935.484375      382.936646       290.0  22.224644   10.173570
3           92581.601562       20.590000       103.0   0.148000    4.625000
4          103012.093750       74.360001       143.0   0.286000    0.429000

   total_protein
0      18.633970
1      25.910593
2      35.345387
3       0.956000
4      20.020000
       mean_depth   min_depth   max_depth  depth_range  relative_volume_score  \
count  300.000000  300.000000  300.000000   300.000000             300.000000
mean     0.384939    0.302173    2.089518     1.787345           95095.195312
std      0.019619    0.026696    2.120949     2.133592            6799.054199
min      0.348627    0.256300    0.416400     0.075100           76447.789062
25%      0.371847    0.278850    0.432350     0.116625           91200.501953
50%      0.383378    0.303100    1.125850     0.801150           96595.109375
75%      0.391357    0.324375    3.396225     3.105050           98961.478516
max      0.485739    0.349700    6.553500     6.296200          108237.710938

       total_calories   total_mass   total_fat  total_carb  total_protein
count      300.000000   300.000000  300.000000  300.000000     300.000000
mean       226.656782   178.340000   11.833113   18.359765      13.821287
std        212.530507   134.966282   14.454809   15.755493      16.755774
min          0.000000     1.000000    0.000000    0.000000       0.000000
25%         58.120000    76.750000    0.386000    6.009289       1.212500
50%        149.501617   138.500000    6.998819   14.482795       5.733000
```

Fig 5: Depth Features and other data

```
Working on: cpu
Epoch [1/5] Train MAE: 195.62 | Test MAE: 161.62
Epoch [2/5] Train MAE: 115.18 | Test MAE: 91.17
Epoch [3/5] Train MAE: 81.09 | Test MAE: 74.26
Epoch [4/5] Train MAE: 74.21 | Test MAE: 78.74
Epoch [5/5] Train MAE: 66.70 | Test MAE: 61.94
Saved model as rgb_depth_calorie_predictor.pth
```

Qualitative Analysis:

Alongside quantitative evaluation, we also performed qualitative experiments with RGB images, ground-truth depth and nutritional predictions. The examples shown below provide further evidence that depth is helpful for nutritional prediction because it explicitly provides information about the relative geometry, structure and size of the food items.

Figure 6 presents a qualitative result with a monocular depth map pretrained on MiDaS. In the RGB food image, there are multiple food groups on the plate, such as fruits, almonds and sausage slices. The associated monocular depth map maintains the relative structure of the food groups as seen in the RGB image. Observe that brighter regions in the depth map correspond to areas that MiDaS estimated to be closer to the camera, or higher from the surface of the food plate, while darker regions correspond to flat areas or areas outside the food plate. From this example, we can see that we can infer valuable depth structure from a single RGB image, without the need for a depth sensor.

In this case (in Figure 6), the true number of calories is (522.61) kcal, whereas our proposed framework estimated (637.80) kcal. Although the predicted value is slightly off the mark, our model still inferred that it has higher calories as it is composed of multiple dense foods and has a larger apparent volume of food. Again, we see that monocular depth estimation provided important geometric cues.

In Figure 7 we present another qualitative example with real RGB food images from Nutrition5k and

true depth maps. In this case, there was less geometry in the RGB image and depth map as there was only a small amount of food on a plate. Here, the true calorie content was (152.64) kcal whereas the estimated value was (109.21) kcal. This error was much smaller than the previous examples with a greater diversity of food items and meals. We believe the improvement was due to the fact that the ground truth depth representation enabled the network to perceive that there was a limited amount of food on the plate and low volume of food.

We also found from these qualitative results that monocular depth estimation was able to provide meaningful information in the form of geometry of a food scene without a depth sensor. We also found that the food variety, table setting and geometric complexity play an important role in performance. Scenes with a greater variety of foods had greater error due to unpredictable geometry and irregular structure of foods.

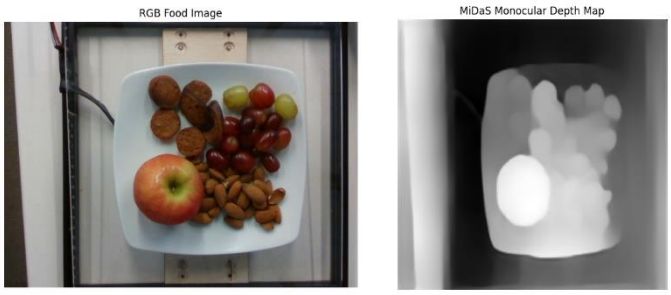


Fig 6: Results with monocular depth

Fig 7: Results without monocular depth

VII. CONCLUSION

Nutri Scan AI is a proposed RGB-depth aware deep learning framework for automatic nutritional value estimation. Automatic feature learning from RGB, encoding of depth with ground truth depth maps, multimodal feature fusion, and monocular depth estimation from RGB images synergistically estimate calories and macros from food images. Testing with the Nutrition5k dataset shows that with depth as an addition to the input, we have a better understanding of food shape, size, and relative food volume than when using only RGB inputs. Our evaluations also found that with MiDaS, a pre-trained monocular depth prediction model, we can predict good quality depth maps from monocular RGB inputs without having to capture depth with other devices. Qualitative estimates of visual, depth, and nutritional values demonstrate that multimodal computer vision can be harnessed for developing smart food diaries. We plan to further this research by adding food segmentation, correct 3D reconstruction of volume, using nutritional values from USDA nutritional data, and by incorporating our algorithm into an interactive mobile app for real time nutritional and health monitoring by the consumer.

REFERENCES

[1] Q. Thames *et al.*, "Nutrition5k: Towards automatic nutritional understanding of generic food," *arXiv [cs.CV]*, 2021.

[2] Nutrition5k: Detailed visual + nutritional data for over 5,000 plates of food. .

[3] N. Karthikeyan, B. Natarajan, S. R. Mahapatro, and D. M. Hussain, "Enhancing dietary management and nutritional analysis through deep learning: a multi-model approach for food classification and volume estimation," Sci. Rep., vol. 16, no. 1, 2026.

AUTHOR CONTRIBUTIONS

1st Author (Sri Varsha Maram): RGB-depth fusion model implementation, data preprocessing, neural network training, MiDaS monocular depth estimation incorporation, experiments

2nd Author (Sri Charan Vutukuru): Nutritional regression pipeline implementation, statistical plots creation, testing depth features, verifying results, helped with report writing.